

Speaker script — Topic 5

SYNC WARNING (2026-08-13). This document describes the earlier **10-slide** cut. `slides.tex` / `slides.pdf` is now the authority: 24 main pages + 5 backup, organised into five sections with an outline and section-divider pages. Content slides present in the PDF but **not** written up here: *The protocol, in the authors' own picture* (paper Fig. 1), *What a classical shadow actually is*, *The benchmark: three spins*, *What each choice actually buys*, *Do you even need shadows?*, *What the shadow inversion actually contributes*, *Which protocol can rebuild which observable?*. Cross-references below use slide **titles**, not numbers, because the numbering has moved.

Written per deck_arc.md script rules: short declarative sentences for non-native English speakers, phonetic hints on hard terms, per-member allocation, timed with slack.

Target: 10 minutes of speech + 10% slack. Times below sum to **8:50**, leaving a 70-second buffer. If you are running long, cut the second bullet of "What we could not do" and backup slides.

Pronunciation hints - *ancilla* — an-SIL-la - *Hadamard* — HA-da-mar (final "d" is silent in French; "HAD-a-mard" is also accepted) - *quadrature* — KWOD-ra-cher - *Trotter* — TROT-ter - *decoherence* — dee-co-HEER-ence - χ (*chi*) — say "kai" (rhymes with "sky") - ρ (*rho*) — say "roe" - *eigenstate* — EYE-gen-state

Numbers rule: every number spoken below appears in RESULTS.md. Do not improvise numbers. If asked something not on a slide, say "that is in our results file, I can show you after."

Slide 1 — Hook (45 s) — Speaker: Chao Hsien

Good morning. We work on the Hadamard test. The Hadamard test is a standard quantum circuit. It uses one extra qubit, called the ancilla. You measure the ancilla. You throw away everything else. We asked one question. What if we do not throw it away? If you measure the discarded qubits in random directions, they become useful data. The same shots now tell you much more. That is our project.

Slide 2 — One record, many quantities (55 s) — Speaker: Jiwan Kang

Here is the circuit. It is small: three system qubits, and one ancilla. We run it once. We keep every shot. From that single record set, we extract seven different physical quantities. The signal χ of t . Three joint observables. And three system observables. The total cost was three thousand four hundred fifty-six circuits, and two hundred fifty-six thousand shots. That is one experiment. Not seven.

Slide 3 — Validation (60 s) — Speaker: Ng Siu Hin

We do not ask you to trust this. We validated everything. The notebook has fifty-six checkpoints. All fifty-six pass. No failures, no warnings. Look at the error bars. Our measured error is zero point zero two nine. Our predicted error bar is zero point zero two eight. The ratio is one point zero five. So our error bars are honest. They are not too small. And the error falls with the square root of the number of shots. The slope is minus zero point four six. Theory says minus zero point five. This means we are limited by statistics, not by a hidden mistake.

Slide 4 — The shared-shot punchline (55 s) — Speaker: Su Wei-Chang

Now the interesting part. Energy and magnetisation are conserved. So they should be flat in time. We measured the energy as zero point zero seven one, plus or minus zero point zero zero eight. The exact value is zero point zero seven four. We measured the magnetisation as two point two seven one. The exact value is two point two six seven. And here is the point. These came from the qubits the standard method throws away. No extra circuits, just a few extra single-qubit gates. There is a statistical cost too, and we quantify it later in the talk.

Slide 5 — MONEY PLOT (75 s) — Speaker: Tina Tien

This is our main result. The grey band is the plain Fourier transform of the ancilla signal. It is a blur. A better estimator sharpens it. But no estimator can colour it. The coloured lines are our reconstruction. We recover all four energy levels. Our worst energy error is zero point zero three three. But look at the colour. The colour is the symmetry sector of each line. That information does not exist in the grey curve. It comes from the recycled data. And it is reproducible. We ran twelve independent random seeds. All twelve got every symmetry label correct. Twelve out of twelve.

(Pause here. This is the slide to let land.)

Slide 6 — Real hardware (70 s) — Speaker: Chao Hsien

We also ran this on real IBM hardware. Our first attempt failed. The signal survived at only eighteen percent. The circuit was too deep. Four hundred thirty-five two-qubit gates. So we built a shallower circuit. One hundred one gates. Same physics. The signal survived at eighty-two percent. On the better machine, eighty-eight percent. And with a full random-basis measurement, we recovered a real system observable on hardware. Ninety-six percent survival. One honest warning. These are single runs. A teammate ran an identical circuit thirty minutes later and got a result two times different. Hardware varies. We report that.

Slide 7 — Honest cost (50 s) — Speaker: Jiwan Kang

Nothing is free. Here is the real cost. To get the same three observables the conventional way, you need thirteen separate experiments. We need zero extra circuits. We do add a few single-qubit gates. And our estimator is noisier on the recycled quantity. We measured that penalty. It is one point zero seven times. Seven percent more noise, to replace thirteen experiments. And we can predict that noise from theory, to within two percent.

Slide 8 — Reproducibility (45 s) — Speaker: Ng Siu Hin

Everything is public and reproducible. Our repository has pinned versions, a summary file you can compare against your own run, and thirty-seven results rows. Every single number maps to the exact command that produced it. You can reproduce our headline numbers in under five minutes, from saved data. If you have quantum hardware access, three commands reproduce our hardware results.

Slide 9 — Limitations (50 s) — Speaker: Su Wei-Chang

We want to be honest about the limits. First. The simple Fourier method never resolves our two closest energy levels. Not at any setting we tried. We needed a

better estimator, not more shots. Second. Our shallow circuit trick only works because our system is small. It gets worse as the system grows. It is not a general recipe. Third. One of our noise studies came out confounded. We excluded it instead of reporting it.

Slide 10 — Team (25 s) — Speaker: Tina Tien

Finally, our team. Chao Hsien opened with the headline scoreboard and the five completed tracks. Jiwan Kang covered the question, the anti-control circuit, and Track B validation. Ng Siu Hin covered classical shadows and the baselines against the echo and HST. Su Wei-Chang covered the AQC compilation win, the phase trap and its fix, and the hardware falsification test. Tina Tien covered Track B on hardware, our claims and scope, and reproducibility. Thank you. We are happy to take questions.

Q&A crib sheet

"How can the compilation trend succeed at $n=7$ if hardware fails harder there?"

They measure different things. The trend is a *ratio* — exact grows like $4^{(n+1)}$, AQC roughly linearly, so the relative advantage grows. Hardware survival depends on the *absolute* depth of the AQC circuit, which still deepens with n : survival $\sim \exp(-\epsilon \times \text{gates})$. Both move at once: relative advantage up, absolute feasibility down. From our own data the effective error is $\sim 6e-3$ per gate at 576 gates; the win becomes visible when that reaches $\sim 2e-3$ — a factor ~ 3 , which is a device-generation step (Nighthawk's target regime), not a protocol fix. [R046, R054]

"Your AQC headline is $n=7$ but the hardware run is $n=6$ — why?"

Different questions. $n=7$ is the last point of the compilation *scaling trend* (gate counts only). The hardware test needs the 8,850-gate exact arm alongside as the doomed *control*, and that arm dominates QPU time — $602 \mu\text{s}$ per shot, about $4\times$ more at $n=7$ on an open-plan quota. The contrast was already saturated at $n=6$ (predicted $3.6e-7$ vs 0.32), so $n=7$ would spend $4\times$ the quantum time making a dead arm deader. And since AQC failed at $n=6$, it would only fail harder at $n=7$ — the choice of n doesn't soften the negative. [R046, R054]

"Is this faster than a classical computer?"

No, and we do not claim that. Our system is three qubits. A laptop solves it instantly. It is small on purpose, so we can check every number against the exact answer.

"What is new here? The paper already exists."

The method is from the paper. What is new is the systematic characterisation. We measured the cost, found the failure modes, compared four spectral estimators -- Fourier, matrix pencil, phase estimation, and Krylov -- plus a protocol ablation, and tested it on three different real devices.

"How do you know your error bars are right?"

We checked them two ways. The measured scatter matches the predicted error bar, ratio one point zero five. And we ran twelve independent seeds and compared the spread to our bootstrap. They agree within the noise of the comparison itself.

"Did anything go wrong?"

Yes, twice, and both are documented. We once applied a shadow estimator to fixed-basis data. That silently measures a different quantity. It was sixty-five sigma wrong

with no noise at all. And we once computed an error bar over pairs instead of over shots, which faked a signal. An ideal-simulator control caught it.

"Can this scale?"

The measurement idea scales. Our specific shallow-circuit trick does not — we show the crossover at four qubits. Beyond that you would need circuit compression, for example tensor-network approximate compiling.

"Is this the first implementation?"

We are not aware of a prior open implementation. We searched on the thirteenth of August and found none. But we scoped that search narrowly, so we would not claim priority on it. The method is from the paper. What is ours is the characterisation.

"What would you do next?"

Density of states. The same circuit, with the input state swapped for a maximally mixed one, measures the density of states instead of one state's spectrum. We prototyped it and it works — the peak heights correctly become degeneracies. There is a recent paper doing exactly this for fermions. It gets one particle-number sector per experiment. Our recycled data could get every sector from a single run, because the same shots give us every observable we need. We have checked that is feasible. We have not built it yet.

"What about the symmetry labels under noise?"

We tested it and the result was confounded, so we are not claiming it. That is honest future work.

Timing summary

slide	time	cumulative
1	0:45	0:45
2	0:55	1:40
3	1:00	2:40
4	0:55	3:35
5	1:15	4:50
6	1:10	6:00
7	0:50	6:50
8	0:45	7:35
9	0:50	8:25
10	0:25	8:50

Total 8:50 — buffer to 10:00. Rehearse once with a timer; non-native speakers should read slower than feels natural.

The 8:50 above is STALE and will overrun — re-time before rehearsing

That table covers 10 content slides. The PDF now has **15** content slides plus an outline and 5 section dividers. The seven not timed above are: *paper Fig. 1*, *what a classical shadow is*, *the benchmark model*, *what each choice buys*, *four ways to treat the garbage*, *what the inversion contributes*, *which protocol rebuilds which observable*.

At the script's own pace that is roughly **+3:30 to +4:00**, i.e. a ~12:30 talk against a 10:00 limit. **Someone must decide what to cut**, not discover it on stage. Cheapest cuts, in order:

1. Drop the section-divider pages (they are \AtBeginSection; deleting that block removes all five at once) — saves ~25 s and 5 pages.
2. Move *what the inversion contributes* and *four ways to treat the garbage* to backup — the observable table already carries the conclusion.
3. Show the paper Fig. 1 slide silently while speaking the hook, rather than narrating it.

Do **not** cut the benchmark-model slide: the observable table on *which protocol rebuilds which observable* refers back to its hopping term.